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11 February 1976
SUPERSEDING
MIL-E-1/1553A(NAVY)
1 July 1970

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 7651

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Tetrode, ceramic-metal
See figure 1
Mounting position: Any
Weight: 12 ounces (340.2 grams) nominal

ABSOLUTE RATINGS: See note 1

Ⓑ

Parameter:	F	Ef	Eb	eb	Ec2	ec2	Ec1
Unit:	MHz	V	Vdc	v	Vdc	v	Vdc
Maximum:							
Pulsed - RF amplifier:	1,215	---	5,000	8,000	1,200	1,200	-250
TEST CONDITIONS:	---	6.3	2,500	---	1,000	---	Vary

ABSOLUTE RATINGS: See note 1

Parameter:	Ib	DC anode current during pulse	Pg2	Pg1	Pp	tk	T(seal)	Barometric pressure, reduced
Unit:	Adc	A	W	W	W	sec	°C	mmHg
Maximum:								
Pulsed - RF amplifier:	0.5	9	25	10	600	120	250	140.7
TEST CONDITIONS:	0.240	---	---	---	---	120	250(max)	---

GENERAL:

Qualification - Required

Ⓑ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1266	Total grid current	4	Ec2 = 400 V	0.65	II	Ic1	---	-50	μAdc
1261	Electrode voltage (1) (grid)	-	Ec2 = 400 Vdc	0.65	II	Ec1	-17	-35	Vdc
1256	Electrode current (screen)	-		0.65	II	Ic2	-22	+2	mAdc
1261	Electrode voltage (2) (grid)	-	Eb = 6,000 Vdc; Ec1/Ib = 5 mAdc	0.65	II	Ec1	---	-185	Vdc
1231	Pulsing emission	4	eb = ec2 = ec1 = 850 v	0.65	II	is	80	---	a
---	Positive grid voltage	6	Eb = 2,000 Vdc; Ec1 = -200 Vdc	0.65	II	ec1	100	300	v
	Quality conformance inspection, part 2								
1266	Primary grid emission (control)	-	Pg1 = 10 watts; t = 300; anode and screen grounded; T(seal) = 100° ± 15° C	---	---	Isc1	---	-40	μAdc
1266	Primary grid emission (screen)	-	Pg2 = 25 watts; t = 300; anode and grid grounded; T(seal) = 100° ± 15° C	---	---	Isc2	---	-40	μAdc
1301	Heater current	-		---	---	If	6.9	8.3	A
1366	Interelement leakage resistance, cold	5	Supply voltage = 200 Vdc; Rs = vary to give full scale deflection with external leads shorted together	---	---	R	8.0	---	Meg
---	Pulsed power output	2	eb = 8,000 v; ec2 = 1,000 v; F = 1,215 ± 60 MHz; t = 120	---	---	po	35	---	kw
(R)1331	Direct-interelectrode capacitance	-	Use shield adapter, RCA Model No. 7506, or equal; all unused electrodes grounded	---	---	Cg1-k	25	31	pF
						Cg1-g2	33	40	pF
						Cg2-p	5.0	8.0	pF
						Cg1-p	---	0.16	pF
						Cp-k	---	0.01	pF
						Cg2-k	---	1.1	pF
	Quality conformance inspection, part 3								
---	Life test	-	Group D; pulsed power output; Ef = 6.3 V (max); t = 500 hours	---	---	---	---	---	---
---	Life-test end point:								
---	Pulsed power output			---	---	po	30	---	kw

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
							MIN	MAX	
	Quality conformance inspection, part 3 -Continued								
1002	Barometric pressure, reduced	12	Pressure-140.7 mmHg; 3,000 Vdc applied between a and k, with k, g1, and g2 tied together	---	---	---	---	---	---
1042	Shock (1), specified pulse	7, 8, 10	Test condition A	---	---	---	---		---
1042	Shock (2), specified pulse	3, 8, 10	Test condition D	---	---	---	---		---
1032	Vibration, mechanical	8, 9, 10	Eb = 300 V; Ec2 - 250 V; R1 = 2,000 ohms; Ec1/Ib = 10 mAdc	---	---	Ep	---	40	Vac

NOTES:

1. Ratings apply for a maximum "on" time of 10 μs in any 1,000 μs interval. Because the cathode is subjected to considerable back bombardment as the frequency is increased with resultant increase in temperature, the heater voltage should be reduced depending on operating conditions and frequency to prevent overheating the cathode and resultant short life.

2. Use cathode drive cavity, 10 μs pulse length, 1 percent duty, driving power = 5 kw. Vary Ec1 to obtain anode current during pulse of 9 amperes. The duty factor shall be measured by viewing the rectified rf pulse. Pulse width shall be measured at 0.707 of average of the top of the rectified rf voltage pulse.

3. The tube shall be subjected to five blows of any sequence in each of these positions: Y+, X+, Z+, and Z- (see figure 2). This is not a destructive test.

4. This test shall be performed at the conclusion of the holding period.

5. The tube shall be cooled (room ambient temperature) 30 minutes before performing this test. The resistance between any two electrodes (except across the heater terminals) shall be measured in both directions.

6. Drive grid positive with a pulse of 8 to 18 μs duration measured at 70.7 percent of the peak value of the pulse. Prr = 60. Adjust positive grid pulse to obtain 28 amperes peak anode current neglecting the overshoot. Positive grid voltage (ec1) is that voltage required to give this peak anode current. The peak value is defined as the maximum value of a smooth curve through the average of the fluctuations over the top portion of the pulse, neglecting the initial overshoot.

7. The tube shall be subjected to 18 impact shocks (see figure 2 for orientation of axes) as follows: Three shocks in each of the mutually perpendicular axes in any sequence: Y+, Y-, X+, X-, Z+, and Z-. This is not a destructive test.

8. There shall be no shorts or opens after this test. Tubes shall meet the limits for the following tests after this test: Total grid current, electrode voltage (1) (grid), and electrode current (screen).

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NOTES: -Continued

9. Vibrate along each of the three mutually perpendicular axes (X, Y, and Z, see figure 2) as follows:

Frequency (Hz)	Double amplitude (inches	Acceleration (G)	Approx sweep duration (minutes)
5-10	0.08 ± 10%		1
10-15		0.41 ± 10%	0.5
15-105	0.036 ± 10%		2.5
105-2,000		20 ± 10%	4

Duration of the cycle shall be at least 8 minutes from 5 to 2,000 Hz. The frequency shall vary approximately logarithmic. This is not a destructive test.

An oscilloscope having a vertical amplifier with a minimum bandpass of dc to 10 MHz, shall be used to monitor the signal across the 2K ohm series anode resistor. The horizontal scope sweep frequency shall be approximately 1 cm per second. (Full scale vertical deflection shall be at least 250 volts.) The signal shall be dc coupled to the oscilloscope. If a signal greater than 60 volts occurs during the vibration sweep, the tube shall be considered a failure for this test. Sixty volts on the scope approximates 40 volts rms on the Bruel and Kjaer recorder, or equivalent.

The 60 volts referred to herein is deflection only, and does not include the oscilloscope signal resulting from the 10 ma anode current flowing through Rp.

Ⓑ 10. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted. Shock and vibration samples may be formulated by combining tube types of the same basic tube type 7650 or 7651 construction. Sample may be composed of either type or a combination of basic generic types.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

User activities:
Army - WC
Navy - AS, OS, MC, CG
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

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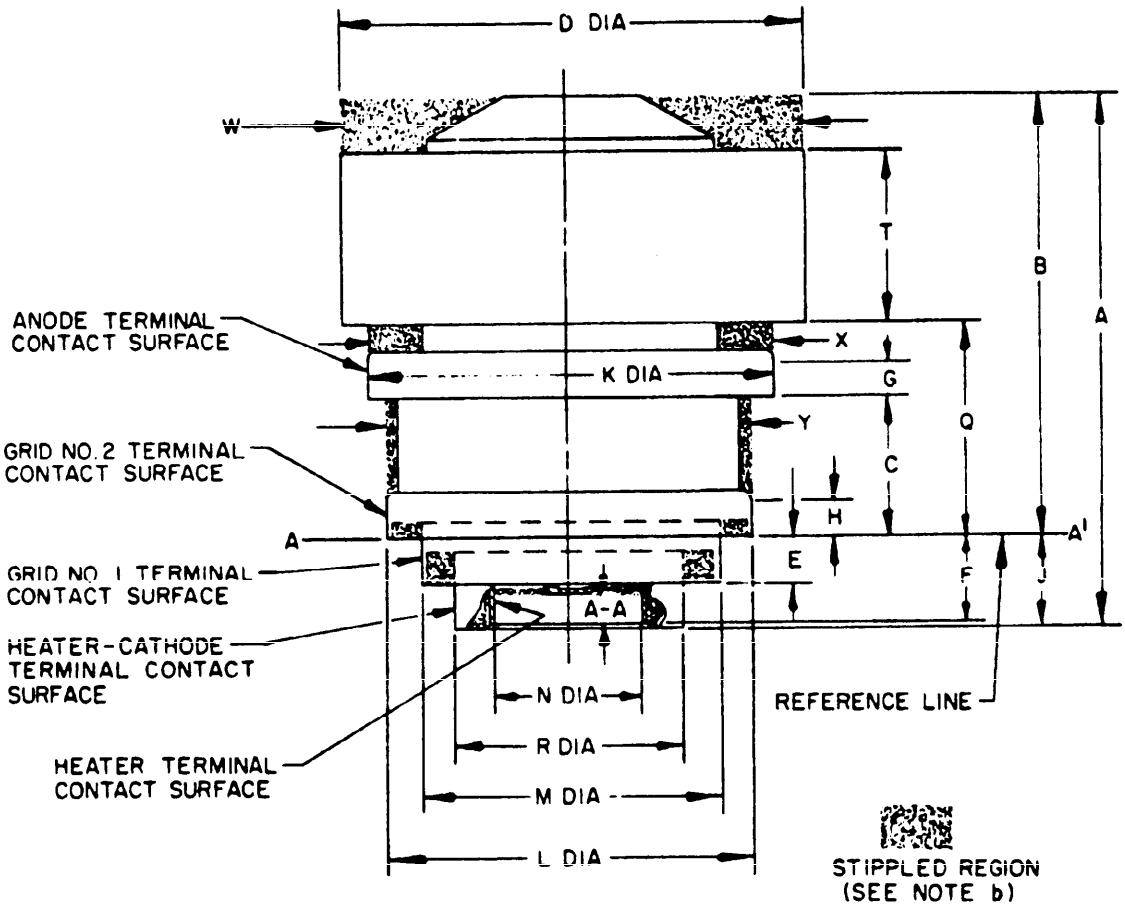


FIGURE 1. Outline drawing of electron tube type 7651.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses		Notes
	Minimum	Maximum	
Quality conformance inspection, part 1			Ⓑ
	Figure 3 - Gage and Note		
A		2.40 (60.96)	
B	1.90 (48.26)	1.98 (50.29)	
C	.550 (13.97)	.600 (15.24)	
D	2.05 (52.07)	2.08 (52.83)	g
E	.18 (4.57)	.22 (5.59)	
J	.38 (9.65)	.42 (10.67)	
K	1.745 (44.32)		f
L	1.59 (40.39)		h
M	1.29 (32.77)		e
N		.67 (17.02)	
Q	.86 (21.84)	.93 (23.62)	
R	.99 (25.15)		c
Quality conformance inspection, part 3			k
F	.360 (9.14)	.410 (10.41)	
G	.145 (3.68)		
H	.15 (3.81)		
T	.735 (18.67)	.775 (19.69)	
AA	.115 (2.92)		
	Reference dimensions		
W	- - -		d
X	- - -		a
Y	- - -		j

NOTES:

a. On any one tube this dimension shall never be greater than K.
b. Keep all stippled regions clear. Do not allow contacts or circuit components to protrude into these annular volumes.
c. Dimension R applies over length J minus E.
d. On any one tube this dimension shall never be greater than D.
e. Dimension M applies over length E only.
f. Dimension K applies over length G only.
g. Dimension D applies over length T only.
h. Dimension L applies over length H only.
j. On any one tube this dimension shall never be greater than L.
Ⓑ k. Dimensions shall be checked during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted. Samples may be formulated by combining tube types of the same basic tube type 7650 or 7651 construction. Samples may be composed of either type or a combination of basic generic types.

FIGURE 1. Outline drawing of electron tube type 7651 - Continued.

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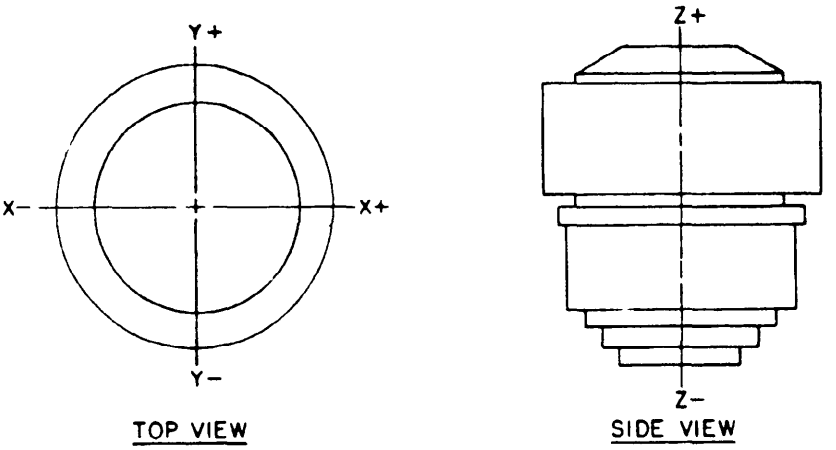
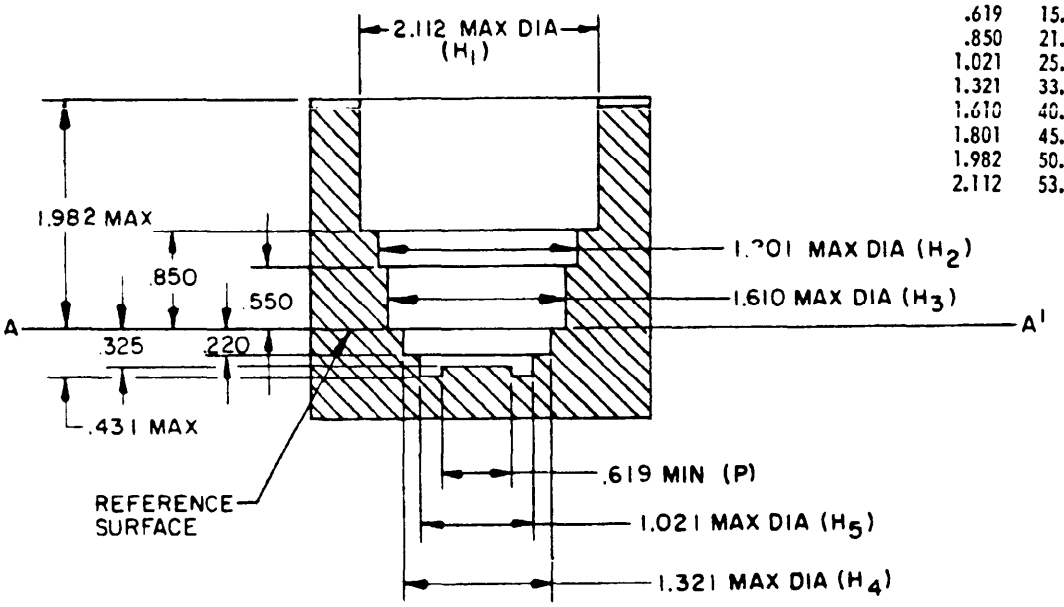


FIGURE 2. Orientation of axes.

INCHES	MM
.220	5.59
.325	8.26
.431	10.95
.550	13.97
.619	15.72
.850	21.59
1.021	25.93
1.321	33.55
1.610	40.89
1.801	45.75
1.982	50.34
2.112	53.64



NOTE:
With the cylindrical surfaces of the radiator band, anode terminal, grid No. 2 terminal, grid No. 1 terminal, heater-cathode terminal, and heater terminal clean, smooth and free of burrs, the tube will enter the gage, to determine out of roundness, concentricity and over size. Proper entry of the tube is obtained when the grid No. 2 terminal is seated on the shoulder marked A-A'. Seating is determined by failure of a .010" thickness gage 1/8" wide to enter more than 1/16" between shoulder surface and grid No. 2 terminal. Slots are provided to permit this measurement to be made.
The axes of cylindrical holes H_1 through H_5 and the axis of the post P shall be coincident within .001". Quality conformance inspection, part 1, AQL 1.0 percent, inspection level II.

FIGURE 3. Gage.

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D.C. 20360

POSTAGE AND FEES PAID
NAVY DEPARTMENT
DOD 316

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